Week 4: Inner products, cross products, and determinants

Linear Algebra I
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These notes were drafted with the help of ChatGPT from my handwritten notes and the discussion used to prepare the recitation.

1. Orthogonal projection onto a line

Last time we studied matrix operations and introduced orthogonal projection onto a line. We begin by writing this projection as a matrix and checking its properties in an example. Orthogonal decomposition then gives Cauchy–Schwarz and explains how to find a closest point in a subspace. We next turn to the cross product and its physical meaning, before studying the definitions of the determinant and selected homework problems.

Vectors are columns. We write $M_{m,n}(\mathbb{R})$ for the real $m \times n$ matrices, $M_n(\mathbb{R}) = M_{n,n}(\mathbb{R})$, I_n for the identity matrix, and A^T for the transpose of A. For $a, b \in \mathbb{R}^n$, the Euclidean inner product and norm are

$$(a \mid b) = \sum_{i=1}^{n} a_i b_i = a^T b = b^T a, \qquad \|a\| = \sqrt{(a \mid a)}.$$

Orthogonality means $(a \mid b) = 0$. The notation $\operatorname{span}(v_1, \ldots, v_m)$ denotes the set of all linear combinations of the indicated vectors.

Given $a \neq 0$ and $u \in \mathbb{R}^n$, we want to write

$$u = p + r, \qquad p \in \mathbb{R}a, \qquad r \perp a.$$

Since $p = ta$ for some scalar t, the perpendicularity condition is

$$0 = (u - ta \mid a) = (u \mid a) - t(a \mid a).$$

Thus the only possible coefficient is $t = (u \mid a)/(a \mid a)$. Conversely, this choice makes the residual perpendicular to a, so the required decomposition exists and is unique:

$$p = \frac{(u \mid a)}{(a \mid a)} a, \qquad r = u - \frac{(u \mid a)}{(a \mid a)} a. \tag{1}$$

Writing the projection as a matrix

The scalar $a^T u$ may be moved past the vector a, and matrix multiplication is associative. Therefore

$$\begin{aligned} p &= \frac{(a^T u)a}{a^T a} = \frac{a(a^T u)}{a^T a} = \frac{(aa^T)u}{a^T a} \\ &= \left(\frac{aa^T}{a^T a}\right) u. \end{aligned}$$

Here $a^T a$ is a positive scalar, while aa^T is an $n \times n$ matrix. Hence

$$\boxed{p = Pu, \qquad P = \frac{aa^T}{a^T a}.} \tag{2}$$

We call P the *orthogonal projection matrix* onto $\mathbb{R}a$.

It is symmetric:

$$P^T = \frac{(aa^T)^T}{a^T a} = \frac{aa^T}{a^T a} = P.$$

It is also idempotent:

$$P^2 = \frac{(aa^T)(aa^T)}{(a^Ta)^2} = \frac{a(a^Ta)a^T}{(a^Ta)^2} = \frac{aa^T}{a^Ta} = P.$$

The identity $P^2 = P$ says that projecting a vector already on the line leaves it fixed.

A numerical example

Take

$$u = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \qquad a = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Then $a^Ta = 5$, $a^Tu = 5$, and

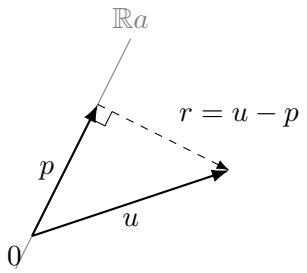
$$P = \frac{1}{5}\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}, \qquad p = Pu = \frac{1}{5}\begin{pmatrix} 5 \\ 10 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

The residual is

$$r = u - p = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \qquad a^Tr = 2 - 2 = 0.$$

For the matrix check,

$$P^2 = \frac{1}{25}\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = \frac{1}{25}\begin{pmatrix} 5 & 10 \\ 10 & 20 \end{pmatrix} = P.$$



2. Cauchy–Schwarz and three examples

The perpendicular decomposition above gives an inequality relating an inner product to the lengths of its inputs. Since $p \perp r$, expanding the square gives

$$\begin{aligned} \|u\|^2 &= \|p + r\|^2 = \|p\|^2 + \|r\|^2 \\ &= \frac{(u \mid a)^2}{\|a\|^2} + \|r\|^2 \geq \frac{(u \mid a)^2}{\|a\|^2}. \end{aligned}$$

Multiplying by $\|a\|^2 > 0$ proves Cauchy–Schwarz:

$$\boxed{|(u \mid a)|^2 \leq \|u\|^2\|a\|^2.} \tag{3}$$

For $a \neq 0$, equality holds precisely when $r = 0$, or equivalently when u is a multiple of a. When $a = 0$, both sides are zero. Thus equality holds exactly when the two vectors are linearly dependent.

Euclidean vectors

For $a = (a_1, \ldots, a_n)^T$ and $b = (b_1, \ldots, b_n)^T$, inequality (3) reads

$$\boxed{\left(\sum_{i=1}^{n} a_ib_i\right)^2 \leq \left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right).}$$

This is the familiar inequality from high school, written as a statement about dot products. The inner-product formulation will allow the same argument to apply to other kinds of vectors.

Continuous functions

Let $\alpha < \beta$, and consider the real vector space $C([\alpha, \beta], \mathbb{R})$ of continuous functions on that interval. Define

$$(f \mid g) = \int_\alpha^\beta f(t)g(t)\,dt, \qquad \|f\| = \left(\int_\alpha^\beta f(t)^2\,dt\right)^{1/2}.$$

This pairing is symmetric and linear in each input. If a continuous function f is nonzero at some point, its absolute value is bounded below by a positive number on a subinterval of positive length. Thus $\int f^2 > 0$, which proves positive definiteness.

The same perpendicular-decomposition proof therefore gives the integral form of Cauchy–Schwarz:

$$\boxed{\left|\int_\alpha^\beta f(t)g(t)\,dt\right|^2 \le \left(\int_\alpha^\beta f(t)^2\,dt\right)\left(\int_\alpha^\beta g(t)^2\,dt\right).}$$

Equality holds exactly when f, g are linearly dependent as functions. In this example the vectors are functions, and the vector space is infinite-dimensional.

Matrices and the Frobenius inner product

The space $M_{m,n}(\mathbb{R})$ can be identified with $\mathbb{R}^{mn}$ by listing the matrix entries. Its corresponding inner product is the *Frobenius inner product*

$$(A \mid B)_F = \operatorname{tr}(A^T B),$$

where tr denotes the sum of the diagonal entries. Indeed, $A^T B$ is an $n \times n$ matrix, and

$$(A^T B)_{jj} = \sum_{i=1}^m a_{ij} b_{ij}.$$

Consequently,

$$\operatorname{tr}(A^T B) = \sum_{j=1}^n \sum_{i=1}^m a_{ij} b_{ij}, \qquad \|A\|_F = \sqrt{\operatorname{tr}(A^T A)} = \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2\right)^{1/2}.$$

Cauchy–Schwarz now gives

$$\boxed{|\operatorname{tr}(A^T B)|^2 \le \operatorname{tr}(A^T A)\operatorname{tr}(B^T B).}$$

Written entirely in entries, this is

$$\left|\sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ij}\right|^2 \le \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2\right)\left(\sum_{i=1}^m \sum_{j=1}^n b_{ij}^2\right).$$

It is ordinary Euclidean Cauchy–Schwarz for the two lists of matrix entries. Equality means that the matrices are linearly dependent as elements of $M_{m,n}(\mathbb{R})$.

3. The least-squares problem

We next extend the closest-point question from a line to a subspace. Let $W = \operatorname{span}(v_1, \ldots, v_m) \subseteq \mathbb{R}^n$. Given $x \in \mathbb{R}^n$, we seek $p \in W$ minimizing $\|x - p\|$. Assume that $v_1, \ldots, v_m$ are linearly independent, and put

$$A = (v_1 \ \cdots \ v_m) \in M_{n,m}(\mathbb{R}).$$

Every point in W can be written uniquely as Ac, with $c \in \mathbb{R}^m$. We first look for a coefficient vector c_0 such that the residual $r = x - Ac_0$ is perpendicular to all columns of A. These conditions are equivalent to

$$A^T(x - Ac_0) = 0, \qquad \text{or} \qquad A^T A c_0 = A^T x.$$

These are the normal equations.

The matrix $A^T A$ is invertible. To see this, if $A^T Ac = 0$, then

$$0 = c^T A^T Ac = \|Ac\|^2,$$

so $Ac = 0$. Independence of the columns gives $c = 0$. Thus the normal equations have the unique solution

$$c_0 = (A^T A)^{-1} A^T x.$$

Set $p = Ac_0$. For any $y \in W$, the vectors $x - p$ and $p - y$ are perpendicular, since the former is perpendicular to W and the latter belongs to W. Hence

$$\|x - y\|^2 = \|(x - p) + (p - y)\|^2 = \|x - p\|^2 + \|p - y\|^2.$$

The minimum is attained exactly when $y = p$. This proves both existence and uniqueness of the closest point.

The minimizing point is therefore

$$\boxed{p = Px, \qquad P = A(A^T A)^{-1} A^T.} \tag{4}$$

The independence assumption is what allows the inverse in this formula. For a one-column matrix $A = a$, it reduces to (2).

4. Inner product spaces

The preceding examples use the same properties of an inner product, even though their vectors are different kinds of objects. We now state those properties explicitly.

Let V be a vector space over $\mathbb{F}$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. An *inner product* is a function

$$(\cdot \mid \cdot) : V \times V \longrightarrow \mathbb{F}$$

satisfying the following conditions for all $x, y, z \in V$ and $\alpha, \beta \in \mathbb{F}$:

(1) Conjugate symmetry:

$$(x \mid y) = \overline{(y \mid x)}.$$

(2) Linearity in the first argument:

$$(\alpha x + \beta y \mid z) = \alpha(x \mid z) + \beta(y \mid z).$$

(3) Positive definiteness: $(x \mid x)$ is a nonnegative real number, and

$$(x \mid x) = 0 \quad \Longleftrightarrow \quad x = 0.$$

For real scalars, conjugate symmetry is ordinary symmetry. For complex scalars, the second argument is conjugate-linear:

$$(x \mid \alpha y + \beta z) = \overline{\alpha}(x \mid y) + \overline{\beta}(x \mid z).$$

The norm induced by the inner product is $\|x\| = \sqrt{(x \mid x)}$.

Cauchy–Schwarz holds in every such space. For $y \neq 0$, put

$$c = \frac{(x \mid y)}{\|y\|^2}, \qquad r = x - cy.$$

Then $(r \mid y) = 0$, and the perpendicular decomposition gives

$$\|x\|^2 = |c|^2\|y\|^2 + \|r\|^2 = \frac{|(x \mid y)|^2}{\|y\|^2} + \|r\|^2.$$

Thus $|(x \mid y)| \le \|x\|\|y\|$, with equality exactly when x, y are linearly dependent. The case $y = 0$ is immediate. The argument uses only the inner-product axioms and applies in finite or infinite dimension.

5. The cross product and its physical meaning

We return to $\mathbb{R}^3$ with its usual inner product and right-handed orientation. For

$$u = (u_1, u_2, u_3)^T, \qquad v = (v_1, v_2, v_3)^T,$$

the cross product is

$$\boxed{u \times v = \begin{pmatrix} u_2v_3 - u_3v_2 \\ u_3v_1 - u_1v_3 \\ u_1v_2 - u_2v_1 \end{pmatrix}.} \tag{5}$$

Substituting into the dot product shows that

$$u^T(u \times v) = v^T(u \times v) = 0.$$

Thus $u \times v$ is perpendicular to both inputs. Expanding its squared length gives

$$\begin{aligned} \|u \times v\|^2 &= (u_2v_3 - u_3v_2)^2 + (u_3v_1 - u_1v_3)^2 + (u_1v_2 - u_2v_1)^2 \\ &= \|u\|^2\|v\|^2 - (u \mid v)^2. \end{aligned}$$

For nonzero u, v with angle θ, it follows that

$$\|u \times v\| = \|u\|\|v\| \sin\theta.$$

This is the area of the parallelogram with edges u, v. A dependent pair gives area zero. The direction is determined by the right-hand rule.

Physical examples

The direction and magnitude of the cross product appear in several basic physical formulas.

Quantity or law	Formula	Meaning
Torque	$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$	The turning effect of a force; magnitude $\|\mathbf{r}\|\|\mathbf{F}\| \sin\theta$.
Angular momentum	$\mathbf{L} = \mathbf{r} \times \mathbf{p}$	Rotational motion about the reference point.
Magnetic Lorentz force	$\mathbf{F}_{\text{mag}} = q\,\mathbf{v} \times \mathbf{B}$	Force perpendicular to the velocity and magnetic field.
Force on a current element	$d\mathbf{F} = I\,d\boldsymbol{\ell} \times \mathbf{B}$	Magnetic force on a small oriented wire segment.

Here $\mathbf{r}$ is position relative to the chosen reference point, $\mathbf{p}$ is momentum, q is charge, $\mathbf{B}$ is the magnetic field, and I is current. The vector $d\boldsymbol{\ell}$ points in the current direction. In particular, the magnetic force on a moving charge is perpendicular to its velocity and therefore does no work.

An example and the algebraic properties

For

$$u = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \qquad v = \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix},$$

we obtain

$$u \times v = \begin{pmatrix} 6 \\ -3 \\ 1 \end{pmatrix}, \qquad \|u \times v\| = \sqrt{36 + 9 + 1} = \sqrt{46}.$$

As a check, $\|u\|^2 = 5$, $\|v\|^2 = 10$, and $(u \mid v) = 2$, giving $5 \cdot 10 - 2^2 = 46$.

The coordinate formula also proves bilinearity and antisymmetry:

$$(\alpha u + \beta w) \times v = \alpha(u \times v) + \beta(w \times v), \qquad u \times v = -v \times u.$$

Linearity in the second argument follows as well. In particular, $u \times u = 0$.

The scalar triple product and orientation

For $u, v, w \in \mathbb{R}^3$, the scalar triple product is

$$(u \times v)^T w = \det \begin{pmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{pmatrix}.$$

The equality follows by expanding the coordinates. If u, v are independent, the base area is $\|u \times v\|$, and the component of w along the unit normal is

$$\frac{(u \times v)^T w}{\|u \times v\|}.$$

Multiplying gives the signed volume. Taking an absolute value gives the ordinary volume:

$$\boxed{\mathrm{Vol}_3(u, v, w) = |(u \times v)^T w|.} \tag{6}$$

If the edges are dependent, both sides are zero. Swapping two edges leaves the ordinary volume unchanged and reverses the orientation, so the signed volume changes sign.

6. Three equivalent definitions of the determinant

In $\mathbb{R}^n$, an ordered list of n edges should determine a signed volume. The properties suggested by volume lead to the following algebraic definition.

Alternating multilinearity and normalization

The determinant is the unique function $D : (\mathbb{R}^n)^n \longrightarrow \mathbb{R}$ with the following properties:

(1) It is linear in each input separately:

$$D(\ldots, \alpha u + \beta v, \ldots) = \alpha D(\ldots, u, \ldots) + \beta D(\ldots, v, \ldots).$$

(2) Swapping two inputs reverses the sign. Equivalently, D vanishes whenever two inputs are equal.

(3) $D(e_1, \ldots, e_n) = 1$.

If $A = (v_1 \ \cdots \ v_n)$ has columns v_j, we write

$$\det A = D(v_1, \ldots, v_n).$$

For a multilinear function over $\mathbb{R}$, the two statements in (2) are equivalent. The swap rule gives $D = -D$ when the two inputs agree, hence $D = 0$. Conversely, expand a zero expression with $u + v$ in both positions; the two mixed terms must sum to zero.

The permutation formula and uniqueness

Write $A = (a_{ij})$, so $v_j = \sum_i a_{ij} e_i$. Multilinearity gives

$$D(v_1, \ldots, v_n) = \sum_{i_1, \ldots, i_n = 1}^{n} a_{i_1,1} \cdots a_{i_n,n} D(e_{i_1}, \ldots, e_{i_n}).$$

Terms with a repeated index vanish. In every remaining term, the index list is a permutation σ of $\{1, \ldots, n\}$, and alternation gives

$$D(e_{\sigma(1)}, \ldots, e_{\sigma(n)}) = \operatorname{sgn}(\sigma).$$

Here S_n denotes the permutations and

$$\operatorname{sgn}(\sigma) = (-1)^{\#\{(i,j):i<j,\ \sigma(i)>\sigma(j)\}}.$$

Consequently, any function satisfying the three properties must have the formula

$$\boxed{\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} \cdots a_{\sigma(n),n}.} \tag{7}$$

This proves uniqueness. Conversely, the right-hand side is linear in each column, changes sign when two columns are exchanged, and equals 1 on I_n. Thus the required function exists. Formula (7) is the second definition of the determinant.

The same expansion gives a fact we will need for matrix products. If F is any alternating multilinear function, without a normalization condition, then

$$\boxed{F(v_1, \ldots, v_n) = \det(v_1, \ldots, v_n) F(e_1, \ldots, e_n).} \tag{8}$$

Thus every such function is a scalar multiple of the determinant.

The recursive definition by minors

Let $A^{(i,j)}$ denote the matrix obtained by deleting row i and column j. With the convention $D_0(\varnothing) = 1$, define recursively

$$D_n(A) = \sum_{j=1}^{n} (-1)^{1+j} a_{1j} D_{n-1}(A^{(1,j)}). \tag{9}$$

In particular, $D_1((a)) = a$. This is the third definition, and it agrees with the permutation formula.

To see this, group the terms of (7) according to the column j in which row 1 is selected, so $\sigma(j) = 1$. The value 1 contributes exactly $j - 1$ inversions, one with each preceding position. Removing that position and renumbering the remaining rows gives the permutation formula for $A^{(1,j)}$, with the additional sign

$$(-1)^{j-1} = (-1)^{1+j}.$$

Summing over j gives (9). The base value and induction then show that the two formulas agree in every size.

Why these properties describe signed volume

Fix the standard orientation of $\mathbb{R}^n$. For an independent ordered base of $n - 1$ edges, let B be its ordinary $(n-1)$-volume and let ν be the unit normal compatible with that order and the orientation. The signed volume after adjoining an edge w is

$$B(\nu \mid w).$$

The inner product is the signed height, so this expression is linear in w. The same argument applies to any edge position. A dependent base gives zero volume. Swapping edges reverses the orientation,

and the standard unit cube has positive volume 1. Thus signed volume satisfies the three defining properties, and uniqueness identifies it with the determinant. Ordinary volume is its absolute value.

7. Multiplicativity of the determinant

We now use the scalar-multiple statement (8) to prove the product formula. Fix $A \in M_n(\mathbb{R})$, and define

$$F(v_1, \dots, v_n) = \det(Av_1, \dots, Av_n).$$

The map $v \mapsto Av$ is linear, so F is multilinear. Equal inputs give equal transformed inputs, so F is alternating as well. By (8),

$$F(v_1, \dots, v_n) = \det(v_1, \dots, v_n) F(e_1, \dots, e_n).$$

If $A = (u_1 \ \cdots \ u_n)$, then $Ae_i = u_i$. Hence

$$F(e_1, \dots, e_n) = \det(Ae_1, \dots, Ae_n) = \det A.$$

Taking $v_1, \dots, v_n$ to be the columns of B, the vectors Av_j are exactly the columns of AB. Therefore

$$\boxed{\det(AB) = \det A \det B.} \tag{10}$$

The argument also covers singular matrices, since no division by a determinant is used.

8. Selected homework problems

The following problems use the definitions and properties above. The numbering refers to Xi Nanhua's *Linear Algebra*.

Exercise 3.2/8: a zero pattern

Use the definition of the determinant to compute

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & 0 & 0 & 0 \\ a_{41} & a_{42} & 0 & 0 & 0 \\ a_{51} & a_{52} & 0 & 0 & 0 \end{vmatrix}.$$

Solution. In a nonzero term of the permutation formula, the entries selected from rows $3, 4, 5$ would all have to lie in columns $1, 2$. But every permutation term selects distinct columns in distinct rows. Three distinct column choices cannot all belong to a set of size two. Thus every term vanishes, and

$$\boxed{\Delta = 0.}$$

Exercise 3.3/2(2)–(3): two Vandermonde identities

Without expanding the determinants, prove

$$(2) \qquad \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix},$$

and

$$(3) \qquad \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (ab + bc + ca) \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}.$$

Solution. Put

$$s_1 = a+b+c, \qquad s_2 = ab+bc+ca, \qquad s_3 = abc, \qquad V = \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}.$$

For (2), perform the column operation

$$C_3 \leftarrow C_3 + s_1 C_2 - s_2 C_1.$$

The first new entry is

$$bc + (a+b+c)a - (ab+bc+ca) = a^2.$$

The other entries are b^2, c^2 by the same calculation. The determinant becomes V, proving (2).

For (3), the identity

$$(t-a)(t-b)(t-c) = t^3 - s_1 t^2 + s_2 t - s_3$$

shows that $t^3 - s_1 t^2 - s_3 = -s_2 t$ for $t = a, b, c$. Thus the operation

$$C_3 \leftarrow C_3 - s_1 C_2 - s_3 C_1$$

gives

$$\begin{aligned} \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} &= \begin{vmatrix} 1 & a^2 & -s_2 a \\ 1 & b^2 & -s_2 b \\ 1 & c^2 & -s_2 c \end{vmatrix} \\ &= -s_2 \begin{vmatrix} 1 & a^2 & a \\ 1 & b^2 & b \\ 1 & c^2 & c \end{vmatrix} = s_2 V. \end{aligned}$$

The last step exchanges columns 2 and 3. Neither proof divides by a difference such as $a-b$, so both include cases in which some variables coincide.

Exercise 3.3/12(1): neighboring-column differences

Compute the n-th order determinant

$$\Delta_n = \begin{vmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ -1 & 0 & 3 & \cdots & n-1 & n \\ -1 & -2 & 0 & \cdots & n-1 & n \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -1 & -2 & -3 & \cdots & 0 & n \\ -1 & -2 & -3 & \cdots & -(n-1) & 0 \end{vmatrix}.$$

Solution. Extract the factor j from column j, obtaining

$$\Delta_n = n! \det K_n, \qquad K_n = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ -1 & 0 & 1 & \cdots & 1 \\ -1 & -1 & 0 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & -1 & -1 & \cdots & 0 \end{pmatrix}.$$

Perform the column operations

$$C_j \leftarrow C_j - C_{j-1}, \qquad j = n, n-1, \ldots, 2,$$

in this order, from right to left. Each predecessor column is then still unchanged when it is used. The first row becomes $(1, 0, \ldots, 0)$. Expanding along that row leaves the $(n-1) \times (n-1)$ upper

triangular determinant

$$\det K_n = \begin{vmatrix} 1 & 1 & 0 & \cdots & 0 \\ 0 & 1 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 1 & 1 \\ 0 & \cdots & \cdots & 0 & 1 \end{vmatrix} = 1.$$

The case $n = 1$ is immediate. Restoring the extracted factors gives

$$\boxed{\Delta_n = n!.}$$

Exercise 3.3/1: divisibility by 29

The integers $1653, 2581, 3451, 4582$ are divisible by 29. Using properties of determinants, rather than calculating its value directly, prove that

$$\Delta = \begin{vmatrix} 1 & 6 & 5 & 3 \\ 2 & 5 & 8 & 1 \\ 3 & 4 & 5 & 1 \\ 4 & 5 & 8 & 2 \end{vmatrix}$$

is divisible by 29.

Solution. Replace the last column by

$$C_4 \leftarrow C_4 + 1000C_1 + 100C_2 + 10C_3.$$

Adding multiples of the other columns leaves the determinant unchanged. Hence

$$\Delta = \begin{vmatrix} 1 & 6 & 5 & 1653 \\ 2 & 5 & 8 & 2581 \\ 3 & 4 & 5 & 3451 \\ 4 & 5 & 8 & 4582 \end{vmatrix} = 29 \begin{vmatrix} 1 & 6 & 5 & 57 \\ 2 & 5 & 8 & 89 \\ 3 & 4 & 5 & 119 \\ 4 & 5 & 8 & 158 \end{vmatrix}.$$

The remaining determinant is an integer because its entries are integers. Therefore $29 \mid \Delta$.